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4 / 4 submitted

Level 2 — 25 marks/question

L2-Q1. rotate an array (M)

C
CODE

```
#include<stdio.h>
int main(){
    int a[5]={1,2,3,4,5},i,temp;
    for(int j=0;j<2;j++){
        temp=a[0];
        for(i=0;i<4;i++){
            a[i]=a[i+1];
        }
        a[4]=temp;
    }
    for(i=0;i<5;i++){
        printf("%d ",a[i]);
    }
    return 0;
}
```

OUTPUT 1 2 3 4 5

+

CODE

```
#include <stdio.h>
#include <math.h>

struct Point
{
    int x;
    int y;
};

int main()
{
    struct Point p;
    float distance;

    printf("Enter x-coordinate: ");
    scanf("%d", &p.x);

    printf("Enter y-coordinate: ");
    scanf("%d", &p.y);

    distance = sqrt(p.x * p.x + p.y * p.y);

    printf("Point Coordinates: (%d, %d)\n", p.x, p.y);
    printf("Distance from origin: %.2f\n", distance);

    if (p.x > 0 && p.y > 0)
        printf("Quadrant: 1\n");
    else if (p.x < 0 && p.y > 0)
        printf("Quadrant: 2\n");
    else if (p.x < 0 && p.y < 0)
        printf("Quadrant: 3\n");
    else if (p.x > 0 && p.y < 0)
        printf("Quadrant: 4\n");
    else
        printf("Point lies on an axis\n");

    return 0;
}
```

OUTPUT (no output)

L2-Q3. convert (M)

+

CODE

```
#include <stdio.h>
int main(){
    char s[100];
    scanf("%s", s);
    for(int i=0; s[i]; i++)
        if(s[i] == 'a' || s[i] == 'e' || s[i] == 'i' || s[i] == 'o' || s[i] == 'u')
            printf("%c", s[i]);
    return 0;
}
```

INPUT aeiio world from space

OUTPUT aeio

1

CODE

```
#include <stdio.h>

int main()
{
    float a[2][2], det;

    printf("Enter elements:\n");
    scanf("%f %f %f %f", &a[0][0], &a[0][1], &a[1][0], &a[1][1]);

    det = a[0][0] * a[1][1] - a[0][1] * a[1][0];

    if (det == 0)
    {
        printf("Inverse does not exist");
    }
    else
    {
        printf("Inverse matrix:\n");
        printf("%.1f %.1f\n", a[1][1] / det, -a[0][1] / det);
        printf("%.1f %.1f\n", -a[1][0] / det, a[0][0] / det);
    }

    return 0;
}
```

OUTPUT

(no output)